Triangle Trouble

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Graphics

1. Objectives

A. To ensure students can initialize WebGL
B. To ensure students understand the difference between
    a. gl.TRIANGLE
    b. gl.TRIANGLE_STRIP
    c. gl.TRIANGLE_FAN

2. Equipment

A. A computer that can run notepad ++ and a modern web browser

3. Lab Preparation

A. Ensure you have the example code working and the WebGL initialized correctly.

4. Lab Instruction

A. You will make one WebGL object to render
B. It will have the following vertices (X,Y), Z will always be 0 and you may choose what colors you want to use for each vertex.
    a. (0,.5)
    b. (1,.5)
    c. (.5,1)
    d. (.5,0)
    e. (0,-1)
    f. (-.5,0)
C. You will also add a variable inside the object that will change how these vertices are rendered
    a. If the variable =
        i. 0 – then use gl.TRIANGLE
        ii. 1 – then use gl.TRIANGLE_STRIP
        iii. 2 – then use gl.TRIANGLE_FAN
D. In the HTML create a simple main class that will have a function which will change this objects variable.
    a. (In the constructor of main instantiate this object)
E. Create either three buttons or a drop down with the three different rendering styles.
F. When the button (or dropdown) is pressed/changed, send the input to a global function, which will then send data to the main class's function.
G. The function inside the main will set the objects variable appropriately
H. Clear the screen with (gl.clear(gl.COLOR_BUFFER_BIT|gl.DEPTH_BUFFER_BIT);)
I. Re-Render the object (with the new mode).

5. Lab Rubric

	Perfect	Close, but not quite right, (no syntax errors)	Not acceptable
Button (or dropdown) works	10	5	0
gl.TRIANGLES renders correctly	5	2	0
gl.TRIANGLE_STRIP renders correctly	5	2	0
gl.TRIANGLE_FAN renders correctly	5	2	0
Total	/25		

6. Lab Report Requirements

No lab report due but there is a peer review required for this assignment.